University of HUDDERSFIELD
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Assignment 2

Course: CHS2406-QGA-S1-2425: Data-driven Artificial Intelligence

Data Acquisition	**Good** 10 points	**Satisfactory** 5 points	**Unsatisfactory** 0 points	**Criterion Score**
Criterion 1	• All images were acquired, resized, renamed, and attached correctly in the corresponding folders, meeting the specifications. • The images are clear and appropriately formatted for the next stages of the task.	• Images submitted were unclear, in unsupported formats, or not named as per the given specifications. • The images were not adequately prepared for further analysis.	• No attempt made to acquire and submit input images.	/ 10

Identifying Label Errors	**Excellent** 20 points	**Very Good** 15 points	**Good** 10 points	**Satisfactory** 5 points	**Unsatisfactory** 0 points	**Criterion Score**
Criterion 1	• Utilised advanced libraries like Cleanlab to accurately identify and report labeling errors with detailed stage-wise counts and error types. • Comprehensive and well-documented analysis with clear justification for the methodology.	• Applied a learning algorithm to predict stage labels and identify errors. • Stage-wise counts and error types are reported with reasonable accuracy and detail.	• Conducted manual verification to identify labeling errors. • Stage-wise counts are provided but lack depth or detailed analysis.	• Some effort made to report labeling errors, but stage-wise details or types are missing or incomplete. • No significant methodology applied to identify errors.	• No attempt made to identify or report labeling errors in the dataset.	/ 20

Preprocessing & data splitting	**Excellent** 10 points	**Very Good** 7 points	**Unsatisfactory** 3 points	**Criterion Score**

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New Criterion	• Preprocessing is comprehensive and includes steps like normalisation and ensures resizing. • Dataset is split correctly into training, validation, and testing sets with an appropriate and justified ratio. • Proper comments were given.	• Preprocessing is partially completed, with some essential steps missing or not well-implemented. • Dataset splitting is done but may not follow an optimal ratio or lacks justification. • Partial comments were given.	• Minimal or no preprocessing performed; critical steps like normalisation or resizing are missing. • Dataset splitting is either incorrect or not attempted. • Little to no comments.	/ 10

Model Implementation & Optimisation	**Excellent** 20 points	**Very Good** 15 points	**Good** 10 points	**Limitted work** 5 points	**Unsatisfactory** 0 points	**Criterion Score**
Model Design and Implementation	• Well-designed CNN architecture tailored to the dataset, with appropriate selection and clear justification of activation functions and optimisation algorithms. • Implementation is error-free, follows good coding practices, and achieves initial performance metrics.	• Functional CNN design with mostly appropriate architecture, activation functions, and optimizers. • Minor gaps in justification or coding quality. • Implementation runs successfully with acceptable performance.	• Basic CNN implemented, but the architecture lacks refinement or relevance to the dataset. • Implementation has minor errors or lacks robustness.	• Attempted a CNN design, but it is incomplete or poorly structured. • Activation functions and optimizers are either missing or incorrect. • Implementation is flawed or does not produce usable results.	• No valid attempt to design or implement a CNN model. • No understanding is demonstrated in the implementation of a CNN model.	/ 20

Model Implementation & Optimisation	**Excellent** 20 points	**Very Good** 15 points	**Good** 10 points	**Limitted work** 5 points	**Unsatisfactory** 0 points	**Criterion Score**
Performance Optimisation and Analysis	• Significant efforts to optimise model performance through advanced techniques such as data augmentation, increasing layers, filters, and hyperparameter tuning. • Comprehensive analysis with comparative results and visualisations (e.g., accuracy/loss curves). • Well-documented insights into what improved performance and why.	• Good attempt at optimising performance with reasonable efforts, including some techniques like data augmentation or architectural changes. • Comparative analysis and visualisations provided but not exhaustive. • Documentation is clear but could go deeper in explaining results.	• Limited attempts at performance improvement; implemented basic changes with little impact on results. • Analysis is minimal, with few or no visualisations. • Documentation is weak or lacks sufficient insights.	• Minimal or superficial attempts at optimisation, with little evidence of performance improvement. • No meaningful analysis or visualisations. • Poor documentation with insufficient detail.	• No valid attempt to optimise the model or analyse performance. • No understanding of optimisation is demonstrated	/ 20

Model Implementation & Optimisation	**Excellent** 20 points	**Very Good** 15 points	**Good** 10 points	**Limitted work** 5 points	**Unsatisfactory** 0 points	**Criterion Score**
Model Evaluation	• Comprehensive evaluation with multiple performance metrics (accuracy, loss, precision, recall, F1-score). • Clear comparison of training vs. validation/test results, with visualisations (accuracy/loss curves) to demonstrate overfitting or underfitting. • In-depth error analysis provided, with clear insights into potential improvements.	• Solid evaluation using key metrics (accuracy, loss) and comparison of training/validation performance. • Visualisations are included but may not fully explain the model's behaviour (e.g., overfitting). • Some error analysis is provided, but it could be more detailed or insightful.	• Basic evaluation with limited metrics (e.g., only accuracy or loss). • Training vs. validation performance is compared but without sufficient analysis or visual aids. • Minimal error analysis with few insights into model performance or areas for improvement.	• Basic or incomplete evaluation, with only one or two metrics used. • Training and validation performance are mentioned but not analysed. • Little to no error analysis, with no visualisations or in-depth examination of results.	• No evaluation performed or reported. • No comparison between training and validation/test performance. • No metrics or error analysis provided.	/ 20

Total	/ 100

Overall Score

Distinction 70 points minimum	**Merit** 60 points minimum	**Satisfactory** 50 points minimum	**Pass** 40 points minimum	**Fail** 0 points minimum